

```
*bui-5503-college-student-data.sas
  author: Hieu Bui
  date: 2026-08-29
  purpose: modify the college student dataset
  license: public domain;

libname perm "/home/u64368263/medb5503/data";

proc import datafile="/home/u64368263/medb5503/data/college student data.sav"
  out=perm.college_data
  dbms=sav
  replace;
run;

proc print data=perm.college_data (obs=10);
run;

proc contents data=perm.college_data;
run;

proc sort data=perm.college_data;
  by gender;
run;

* descriptive analysis by gender;
proc means data=perm.college_data;
  by gender;
  var height hrstv hrsstudy;
run;

* proc reg with ods predict height with pheight;
proc reg data=perm.college_data;
  by gender;
  model height=pheight;
  ods output ANOVA=anovatable;
  title1 "Coefficient of parent's height is 0.48
  which every height unit of parent, the child's predicted height
  increases by 0.47982 units.";
  title2 "The relationship between same-sex parent's height
  and child's height is statistically significant.";
run;

proc print data=anovatable;
run;

* Run a logistic regression in SAS with "television shows-sitcoms"
as outcome variable and use age and gender to predict the outcome variable;
proc logistic data=perm.college_data descending;
  ods output Parameterestimates = coeff;
```

```
ods output Oddsratios = odds_ratios;
model tvsitcom=age gender;
run;

proc print data= coeff;
  title1 "For each one year increase in age, the odds of
  watching sitcoms decreases by 0.69. This is not significant.";
  title2 "For females compared to males, the odds of
  watching sitcoms increases by 2.60. This is not significant.";
run;

proc print data=odds_ratios;
  title1 "Females have 13.44 times higher odds
  of watching sitcoms than males.";
run;

* Is there a difference in GPA between male and female students?;
proc ttest data=perm.college_data;
  class gender;
  var curr GPA;
  title1 "Females GPA is 0.31 points higher than males.";
run;
```

```

1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
NOTE: ODS statements in the SAS Studio environment may disable some output features.
69
70      *bui-5503-college-student-data.sas
71      author: Hieu Bui
72      date: 2026-08-29
73      purpose: modify the college student dataset
74      license: public domain;
75
76      libname perm "/home/u64368263/medb5503/data";
NOTE: Libref PERM was successfully assigned as follows:
Engine:          V9
Physical Name:   /home/u64368263/medb5503/data
77
78      proc import datafile="/home/u64368263/medb5503/data/college student data.sav"
79          out=perm.college_data
80          dbms=sav
81          replace;
82      run;

```

NOTE: The import data set has 50 observations and 18 variables.

NOTE: PERM.COLLEGE_DATA data set was successfully created.

NOTE: PROCEDURE IMPORT used (Total process time):

real time	0.01 seconds	
user cpu time	0.00 seconds	
system cpu time	0.00 seconds	
memory	2245.43k	
OS Memory	27496.00k	
Timestamp	08/30/2026 05:06:23 AM	
Step Count	288	Switch Count 2
Page Faults	0	
Page Reclaims	454	
Page Swaps	0	
Voluntary Context Switches	43	
Involuntary Context Switches	0	
Block Input Operations	0	
Block Output Operations	312	

```

83
84      proc print data=perm.college_data (obs=10);
85      run;

```

NOTE: There were 10 observations read from the data set PERM.COLLEGE_DATA.

NOTE: PROCEDURE PRINT used (Total process time):

real time	0.02 seconds	
user cpu time	0.03 seconds	
system cpu time	0.00 seconds	
memory	1339.65k	
OS Memory	26020.00k	
Timestamp	08/30/2026 05:06:23 AM	
Step Count	289	Switch Count 0
Page Faults	0	
Page Reclaims	61	
Page Swaps	0	

Voluntary Context Switches	9
Involuntary Context Switches	1
Block Input Operations	0
Block Output Operations	8

```

86
87      proc contents data=perm.college_data;
88      run;

```

NOTE: PROCEDURE CONTENTS used (Total process time):

real time	0.03 seconds
user cpu time	0.03 seconds
system cpu time	0.00 seconds
memory	1296.43k
OS Memory	26280.00k
Timestamp	08/30/2026 05:06:23 AM
Step Count	290
Switch Count	0
Page Faults	0
Page Reclaims	91
Page Swaps	0
Voluntary Context Switches	9
Involuntary Context Switches	1
Block Input Operations	0
Block Output Operations	24

```

89
90      proc sort data=perm.college_data;
91      by gender;
92      run;

```

NOTE: There were 50 observations read from the data set PERM.COLLEGE_DATA.

NOTE: The data set PERM.COLLEGE_DATA has 50 observations and 18 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time	0.01 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	985.00k
OS Memory	26280.00k
Timestamp	08/30/2026 05:06:23 AM
Step Count	291
Switch Count	1
Page Faults	0
Page Reclaims	113
Page Swaps	0
Voluntary Context Switches	39
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	264

```

93
94      * descriptive analysis by gender;
95      proc means data=perm.college_data;
96      by gender;
97      var height hrstv hrsstudy;
98      run;

```

NOTE: There were 50 observations read from the data set PERM.COLLEGE_DATA.

NOTE: PROCEDURE MEANS used (Total process time):

real time	0.03 seconds	
user cpu time	0.03 seconds	
system cpu time	0.00 seconds	
memory	1941.21k	
OS Memory	27048.00k	
Timestamp	08/30/2026 05:06:23 AM	
Step Count	292	Switch Count 2
Page Faults	0	
Page Reclaims	180	
Page Swaps	0	
Voluntary Context Switches	19	
Involuntary Context Switches	1	
Block Input Operations	0	
Block Output Operations	16	

```

99
100      * proc reg with ods predict height with pheight;
101      proc reg data=perm.college_data;
102      by gender;
103      model height=pheight;
104      ods output ANOVA=anovatable;
105      title1 "Coefficient of parent's height is 0.48
106      which every height unit of parent, the child's predicted height
107      increases by 0.47982 units.";
108      title2 "The relationship between same-sex parent's height
109      and child's height is statistically significant.";
110      run;

```

NOTE: Interactivity disabled with BY processing.

NOTE: PROCEDURE REG used (Total process time):

real time	0.53 seconds	
user cpu time	0.28 seconds	
system cpu time	0.06 seconds	
memory	19784.21k	
OS Memory	43236.00k	
Timestamp	08/30/2026 05:06:24 AM	
Step Count	293	Switch Count 47
Page Faults	0	
Page Reclaims	26751	
Page Swaps	0	
Voluntary Context Switches	1671	
Involuntary Context Switches	12	
Block Input Operations	0	
Block Output Operations	2208	

NOTE: The data set WORK.ANOVATABLE has 6 observations and 9 variables.

```

111
112      proc print data=anovatable;
113      run;

```

NOTE: There were 6 observations read from the data set WORK.ANOVATABLE.

NOTE: PROCEDURE PRINT used (Total process time):

```

real time          0.01 seconds
user cpu time      0.01 seconds
system cpu time    0.00 seconds
memory            681.21k
OS Memory          34980.00k
Timestamp          08/30/2026 05:06:24 AM
Step Count         294   Switch Count   0
Page Faults        0
Page Reclaims      61
Page Swaps         0
Voluntary Context Switches 0
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 0

```

```

114
115      * Run a logistic regression in SAS with "television shows-sitcoms"
116      as outcome variable and use age and gender to predict the outcome variable;
117      proc logistic data=perm.college_data descending;
118      ods output Parameterestimates = coeff;
119      ods output Oddsratios = odds_ratios;
120      model tvsitcom=age gender;
121      run;

```

NOTE: PROC LOGISTIC is modeling the probability that tvsitcom='yes'.

NOTE: Convergence criterion (GCONV=1E-8) satisfied.

NOTE: The data set WORK.ODDS_RATIOS has 2 observations and 4 variables.

NOTE: The data set WORK.COEFF has 3 observations and 7 variables.

NOTE: There were 50 observations read from the data set PERM.COLLEGE_DATA.

NOTE: PROCEDURE LOGISTIC used (Total process time):

```

real time          0.05 seconds
user cpu time      0.05 seconds
system cpu time    0.00 seconds
memory            2867.90k
OS Memory          36540.00k
Timestamp          08/30/2026 05:06:24 AM
Step Count         295   Switch Count   6
Page Faults        0
Page Reclaims      385
Page Swaps         0
Voluntary Context Switches 51
Involuntary Context Switches 2
Block Input Operations 0
Block Output Operations 576

```

```

122
123      proc print data= coeff;
124      title1 "For each one year increase in age, the odds of
125      watching sitcoms decreases by 0.69. This is not significant.";
126      title2 "For females compared to males, the odds of
127      watching sitcoms increases by 2.60. This is not significant.";
128      run;

```

NOTE: There were 3 observations read from the data set WORK.COEFF.

NOTE: PROCEDURE PRINT used (Total process time):

```

real time          0.01 seconds
user cpu time      0.01 seconds
system cpu time    0.00 seconds
memory            727.75k
OS Memory          34980.00k
Timestamp          08/30/2026 05:06:24 AM
Step Count         296   Switch Count   2
Page Faults        0
Page Reclaims      68
Page Swaps         0
Voluntary Context Switches 20
Involuntary Context Switches 0
Block Input Operations 0
Block Output Operations 8

```

```

129
130     proc print data=odds_ratios;
131     title1 "Females have 13.44 times higher odds
132     of watching sitcoms than males.";
133     run;

```

NOTE: There were 2 observations read from the data set WORK.ODDS_RATIOS.

NOTE: PROCEDURE PRINT used (Total process time):

```

real time          0.00 seconds
user cpu time      0.00 seconds
system cpu time    0.00 seconds
memory            614.93k
OS Memory          34980.00k
Timestamp          08/30/2026 05:06:24 AM
Step Count         297   Switch Count   1
Page Faults        0
Page Reclaims      61
Page Swaps         0
Voluntary Context Switches 9
Involuntary Context Switches 1
Block Input Operations 0
Block Output Operations 0

```

```

134
135     * Is there a difference in GPA between male and female students?;
136     proc ttest data=perm.college_data;
137     class gender;
138     var curr GPA;
139     title1 "Females GPA is 0.31 points higher than males.";
140     run;

```

NOTE: PROCEDURE TTEST used (Total process time):

```

real time          0.22 seconds
user cpu time      0.12 seconds
system cpu time    0.04 seconds
memory            9440.46k
OS Memory          41416.00k
Timestamp          08/30/2026 05:06:24 AM
Step Count         298   Switch Count  49
Page Faults        0

```

Page Reclaims	26219
Page Swaps	0
Voluntary Context Switches	1003
Involuntary Context Switches	6
Block Input Operations	0
Block Output Operations	1096

141

142

143 OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;

153

Obs	height	pheight	gender	marital	age	children	hrstv	tvstitcom	tvmmovies	tvsports	tvnews	hrsstudy	currghpa	evalinst	evalprog	evalphys	evalsocl	hrswork
1	67.00	66.00	females		less than 22	no	18	yes	yes	no	no	5	3.2	neutral	neutral	disagree	agree	0
2	72.00	72.00	males	married	30 or more	yes	4	no	no	yes	no	24	2.6	neutral		neutral	agree	43
3	61.00	62.00	females	divorced	30 or more	yes	25	yes	yes	no	no	20	3.0	disagree	disagree	disagree	disagree	.
4	71.00	71.00	males	single	22-29	no	6	no	no	yes	yes	17	3.1	neutral	disagree	neutral	neutral	37
5	65.00	63.00	females	single	less than 22	no	8	yes	yes	no	yes	18	3.0	neutral	neutral	disagree	disagree	35
6	67.00	68.00	males	single	22-29	no	10	no	no	yes	no	10	3.0	neutral	agree	neutral	disagree	38
7	69.00	74.00	males	single	22-29	no	24	yes	no	yes	yes	12	3.0	disagree	disagree	disagree	strongly disagree	0
8	75.00	73.00	males	married	30 or more	yes	14	no	no	yes	no	10	3.6	neutral	neutral	neutral	strongly disagree	20
9	62.00	65.00	females	divorced	30 or more	no	5	yes	yes	no	yes	8	3.4	neutral	disagree	neutral	disagree	40
10	61.00	64.00	females	married	30 or more	yes	5	yes	yes	no	yes	17	3.5	agree	agree	agree	neutral	38

The CONTENTS Procedure

Data Set Name	PERM.COLLEGE_DATA	Observations	50
Member Type	DATA	Variables	18
Engine	V9	Indexes	0
Created	08/30/2026 00:06:24	Observation Length	144
Last Modified	08/30/2026 00:06:24	Deleted Observations	0
Protection		Compressed	NO
Data Set Type		Sorted	NO
Label			
Data Representation	SOLARIS_X86_64, LINUX_X86_64, ALPHA_TRU64, LINUX_IA64		
Encoding	utf-8 Unicode (UTF-8)		

Engine/Host Dependent Information

Data Set Page Size	131072
Number of Data Set Pages	1
First Data Page	1
Max Obs per Page	909
Obs in First Data Page	50
Number of Data Set Repairs	0
Filename	/home/u64368263/medb5503/data/college_data.sas7bdat
Release Created	9.0401M8
Host Created	Linux
Inode Number	18054998008
Access Permission	rw-r--r--
Owner Name	u64368263
File Size	256KB
File Size (bytes)	262144

Alphabetic List of Variables and Attributes

#	Variable	Type	Len	Format	Label
5	age	Num	8	AGEA.	age group
6	children	Num	8	CHILDREA.	does subject have children
13	currghpa	Num	8	F8.1	student's current gpa
14	evalinst	Num	8	EVALINSA.	positive evaluation, institution
16	evalphys	Num	8	EVALPHYA.	positive evaluation, facilities
15	evalprog	Num	8	EVALPROA.	positive evaluation, major
17	evalsocl	Num	8	EVALSOCA.	positive eval, social life

3	gender	Num	8	GENDERA.	gender of student
1	height	Num	8	F8.2	student height in inches
12	hrsstudy	Num	8	F8.	hours of study per week
7	hrstv	Num	8	F8.	amount of tv watched per week
18	hrswork	Num	8	F8.	hours per week spent working
4	marital	Num	8	MARITALA.	marital status
2	pheight	Num	8	F8.2	same sex parent's height
9	tvmovies	Num	8	TVMOVIEA.	television shows-movies
11	tvnews	Num	8	TVNEWSA.	television shows-news shows
8	tvsitcom	Num	8	TVSITCOA.	television shows-sitcoms
10	tvsports	Num	8	TVSPORTA.	television shows-sports

The MEANS Procedure

gender of student=males

Variable	Label	N	Mean	Std Dev	Minimum	Maximum
height	student height in inches	26	70.2307692	2.8043922	64.0000000	75.0000000
hrstv	amount of tv watched per week	26	13.0384615	6.3275952	4.0000000	24.0000000
hrsstudy	hours of study per week	26	13.3461538	7.2715462	2.0000000	35.0000000

gender of student=females

Variable	Label	N	Mean	Std Dev	Minimum	Maximum
height	student height in inches	24	64.1250000	2.0707592	60.0000000	68.0000000
hrstv	amount of tv watched per week	24	10.8333333	5.7458239	4.0000000	25.0000000
hrsstudy	hours of study per week	24	18.0833333	8.8017620	5.0000000	38.0000000

Coefficient of parent's height is 0.48 which every height unit of parent, the child's predicted height increases by 0.47982 units.
The relationship between same-sex parent's height and child's height is statistically significant.

The REG Procedure

Model: MODEL1

Dependent Variable: height student height in inches

gender of student=males

Number of Observations Read	26
Number of Observations Used	26

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	87.10598	87.10598	19.09	0.0002
Error	24	109.50940	4.56289		
Corrected Total	25	196.61538			

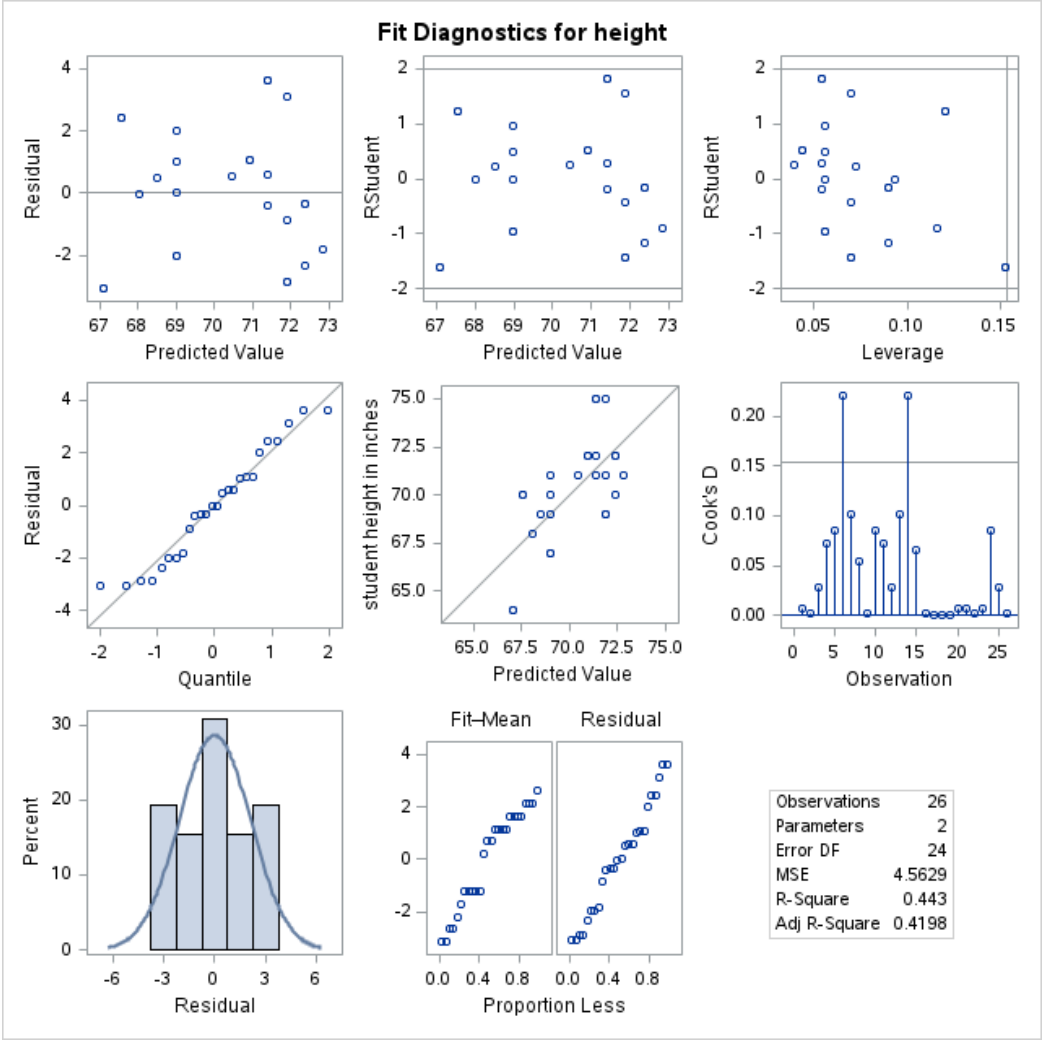
Root MSE	2.13609	R-Square	0.4430
Dependent Mean	70.23077	Adj R-Sq	0.4198
Coeff Var	3.04153		

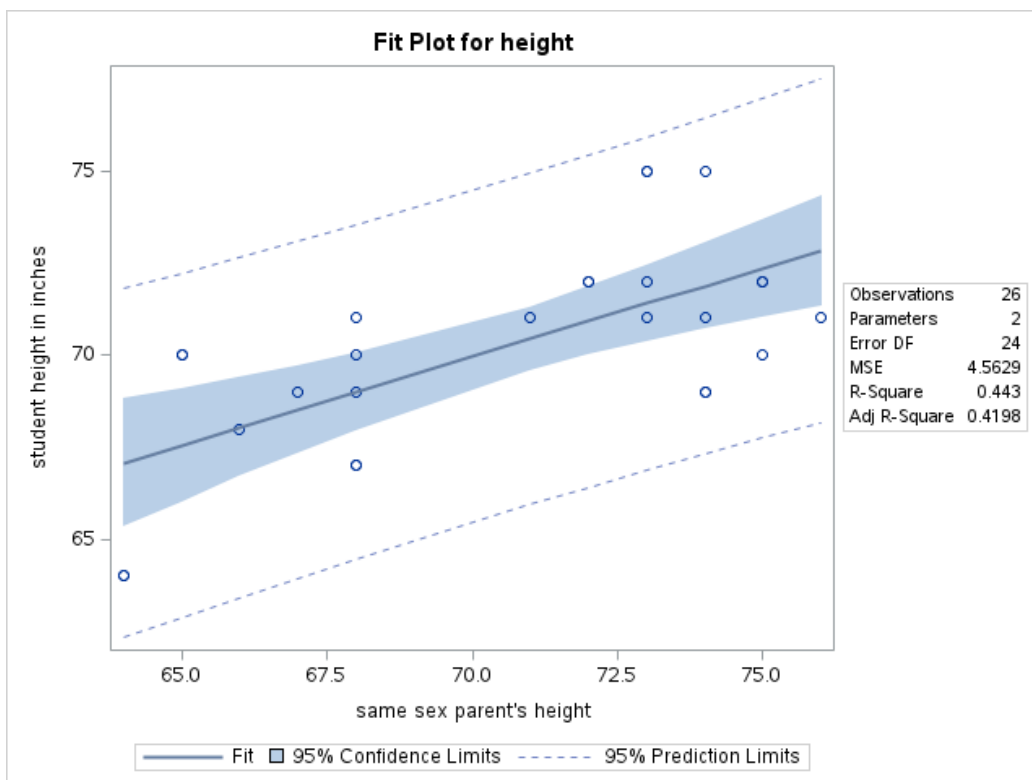
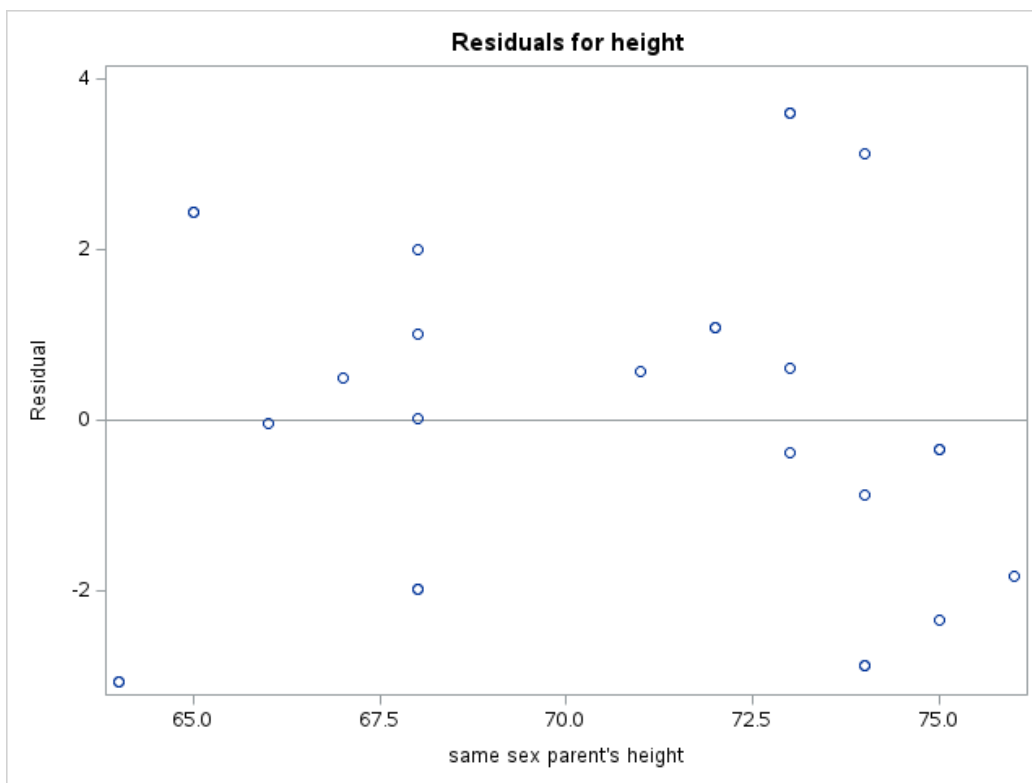
Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	36.36647	7.76196	4.69	<.0001
pheight	same sex parent's height	1	0.47982	0.10982	4.37	0.0002

Coefficient of parent's height is 0.48 which every height unit of parent, the child's predicted height increases by 0.47982 units.
The relationship between same-sex parent's height and child's height is statistically significant.

The REG Procedure

Model: MODEL1
Dependent Variable: height student height in inches
gender of student=males





**Coefficient of parent's height is 0.48 which every height unit of parent, the child's predicted height increases by 0.47982 units.
The relationship between same-sex parent's height and child's height is statistically significant.**

The REG Procedure
Model: MODEL1
Dependent Variable: height student height in inches

gender of student=females

Number of Observations Read	24
Number of Observations Used	24

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	17.25852	17.25852	4.67	0.0419
Error	22	81.36648	3.69848		
Corrected Total	23	98.62500			

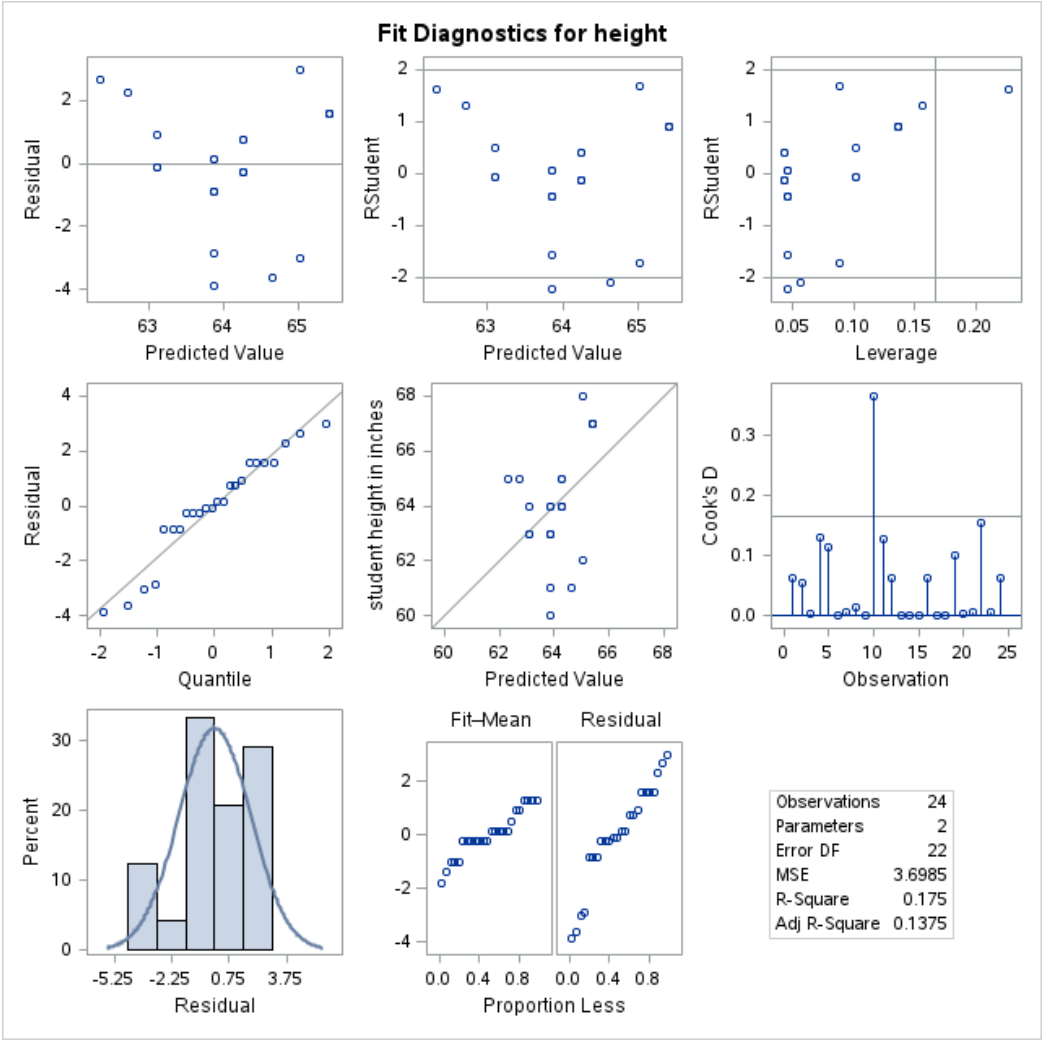
Root MSE	1.92314	R-Square	0.1750
Dependent Mean	64.12500	Adj R-Sq	0.1375
Coeff Var	2.99905		

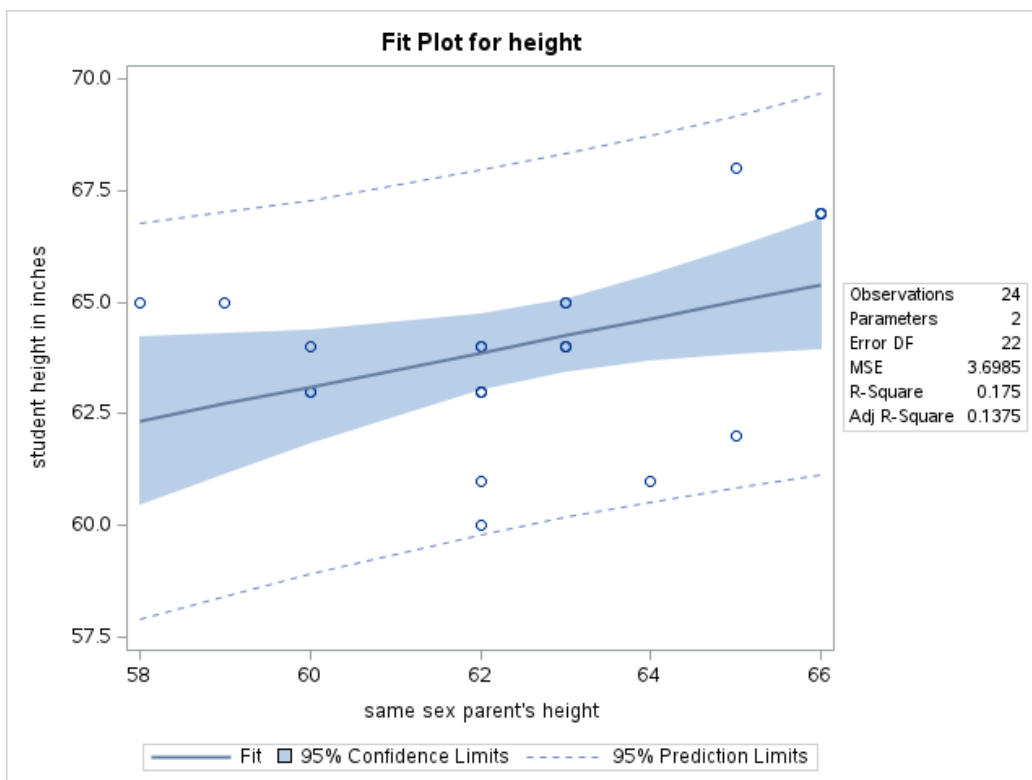
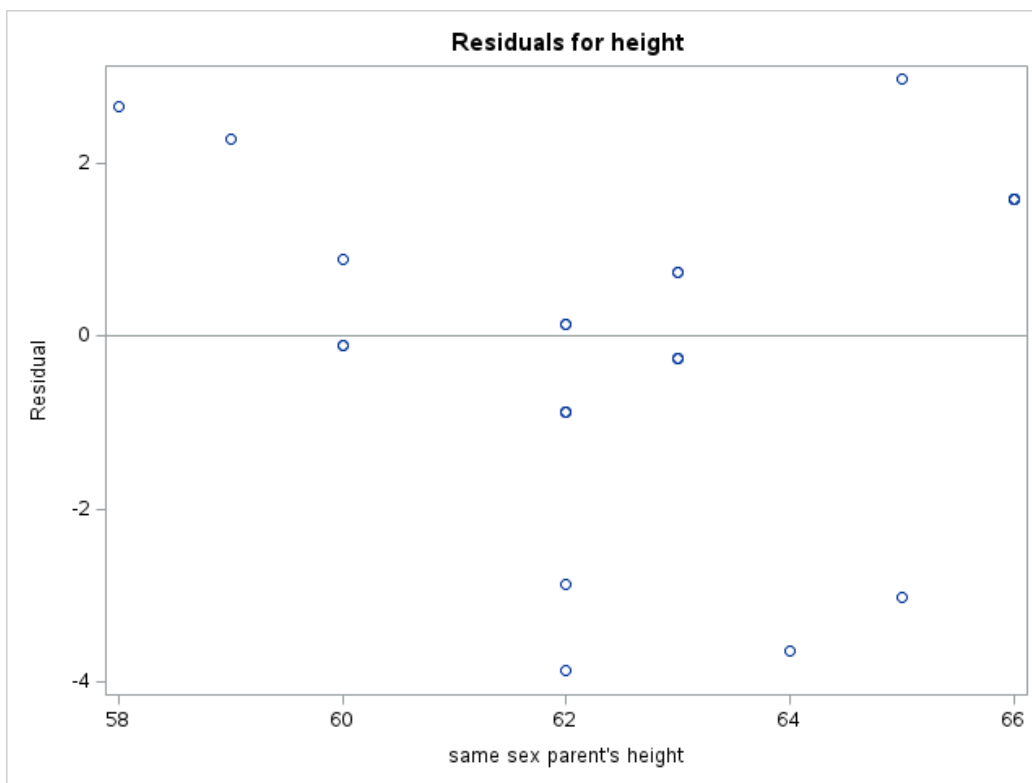
Parameter Estimates						
Variable	Label	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	Intercept	1	40.09091	11.13288	3.60	0.0016
pheight	same sex parent's height	1	0.38352	0.17754	2.16	0.0419

Coefficient of parent's height is 0.48 which every height unit of parent, the child's predicted height increases by 0.47982 units.
The relationship between same-sex parent's height and child's height is statistically significant.

The REG Procedure
Model: MODEL1
Dependent Variable: height student height in inches

gender of student=females





Coefficient of parent's height is 0.48 which every height unit of parent, the child's predicted height increases by 0.47982 units.
The relationship between same-sex parent's height and child's height is statistically significant.

Obs	gender	Model	Dependent	Source	DF	SS	MS	FValue	ProbF
1	males	MODEL1	height	Model	1	87.10598	87.10598	19.09	0.0002

2	males	MODEL1	height	Error	24	109.50940	4.56289	—	—
3	males	MODEL1	height	Corrected Total	25	196.61538	—	—	—
4	females	MODEL1	height	Model	1	17.25852	17.25852	4.67	0.0419
5	females	MODEL1	height	Error	22	81.36648	3.69848	—	—
6	females	MODEL1	height	Corrected Total	23	98.62500	—	—	—

**Coefficient of parent's height is 0.48 which every height unit of parent, the child's predicted height increases by 0.47982 units.
The relationship between same-sex parent's height and child's height is statistically significant.**

The LOGISTIC Procedure

Model Information		
Data Set	PERM.COLLEGE_DATA	
Response Variable	tvsitcom	television shows-sitcoms
Number of Response Levels	2	
Model	binary logit	
Optimization Technique	Fisher's scoring	

Number of Observations Read	50
Number of Observations Used	50

Response Profile		
Ordered Value	tvsitcom	Total Frequency
1	yes	32
2	no	18

Probability modeled is tvsitcom='yes'.

Model Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Intercept Only	Intercept and Covariates
AIC	67.342	57.196
SC	69.254	62.932
-2 Log L	65.342	51.196

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	14.1457	2	0.0008
Score	12.8062	2	0.0017
Wald	10.2639	2	0.0059

Analysis of Maximum Likelihood Estimates					
Parameter	DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept	1	-1.7068	1.1387	2.2468	0.1339
age	1	-0.6790	0.4643	2.1390	0.1436
gender	1	2.5986	0.8131	10.2129	0.0014

Odds Ratio Estimates			
Effect	Point Estimate	95% Wald Confidence Limits	
age	0.507	0.204	1.260
gender	13.444	2.732	66.172

Association of Predicted Probabilities and Observed Responses			
Percent Concordant	72.9	Somers' D	0.592
Percent Discordant	13.7	Gamma	0.683

Percent Tied	13.4	Tau-a	0.278
Pairs	576	c	0.796

For each one year increase in age, the odds of watching sitcoms decreases by 0.69. This is not significant.
For females compared to males, the odds of watching sitcoms increases by 2.60. This is not significant.

Obs	Variable	DF	Estimate	StdErr	WaldChiSq	ProbChiSq	_ESTTYPE_
1	Intercept	1	-1.7068	1.1387	2.2468	0.1339	MLE
2	age	1	-0.6790	0.4643	2.1390	0.1436	MLE
3	gender	1	2.5986	0.8131	10.2129	0.0014	MLE

Females have 13.44 times higher odds of watching sitcoms than males.

Obs	Effect	OddsRatioEst	LowerCL	UpperCL
1	age	0.507	0.204	1.260
2	gender	13.444	2.732	66.172

Females GPA is 0.31 points higher than males.

The TTEST Procedure

Variable: curr GPA (student's current GPA)

gender	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
males		26	3.0231	0.3983	0.0781	2.4000	3.8000
females		24	3.3333	0.3171	0.0647	3.0000	4.0000
Diff (1-2)	Pooled		-0.3103	0.3617	0.1024		
Diff (1-2)	Satterthwaite		-0.3103		0.1015		

gender	Method	Mean	95% CL Mean	Std Dev	95% CL Std Dev
males		3.0231	2.8622 3.1840	0.3983	0.3124 0.5498
females		3.3333	3.1994 3.4673	0.3171	0.2465 0.4449
Diff (1-2)	Pooled	-0.3103	-0.5161 -0.1044	0.3617	0.3016 0.4519
Diff (1-2)	Satterthwaite	-0.3103	-0.5143 -0.1062		

Method	Variances	DF	t Value	Pr > t
Pooled	Equal	48	-3.03	0.0039
Satterthwaite	Unequal	47.023	-3.06	0.0037

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	25	23	1.58	0.2757

